Digit Classification

Random Forest vs. Support Vector Machine — Performance Evaluation Report

1. Overview

Two supervised classifiers: a Random Forest ensemble and a Support Vector Machine (SVM), were trained and evaluated on an optical handwritten digit recognition task. Each sample is an 8×8 grayscale image (64 pixel-intensity features) labeled with a digit class from 0 to 9. Hyperparameters were optimized using grid search with cross-validation on the training set, and final performance was measured on a held-out test set of 360 samples (a 20% split) that neither model saw during tuning

2. Random Forest Hyperparameter Tuning

Grid search identified the following configuration as optimal, achieving a mean cross-validation accuracy of 0.9756:

Parameter	Selected Value
n_estimators	200
max_depth	10
min_samples_split	2
min_samples_leaf	1
Cross-validation accuracy	0.9756

3. Test Set Performance

Both models generalize well to unseen data. The table and chart below summarize accuracy, precision, recall, and F1-score (macro-averaged across all 10-digit classes) on the test set

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9639	0.9647	0.9636	0.9635
SVM	0.9750	0.9760	0.9748	0.9748

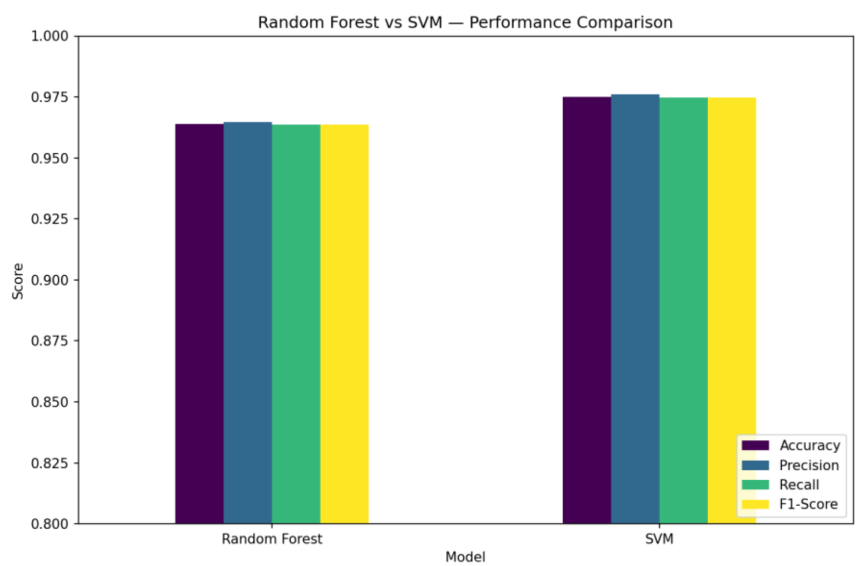

Figure 1. Test-set performance comparison — Random Forest vs. SVM.

The SVM edges out the Random Forest on every metric, with roughly a 1.1 percentage-point advantage in accuracy (97.50% vs. 96.39%) and similarly consistent gains in precision, recall, and F1-score. The narrow gap and the closeness of precision/recall within each model indicate that neither classifier is systematically biased toward over- or under-predicting any particular class

4. Error Analysis: Confusion Matrix & Feature Importance

The Random Forest confusion matrix (left) shows true vs. predicted labels on the test set; the heatmap (right) maps the model's per-pixel feature importances back onto the original 8×8 image grid

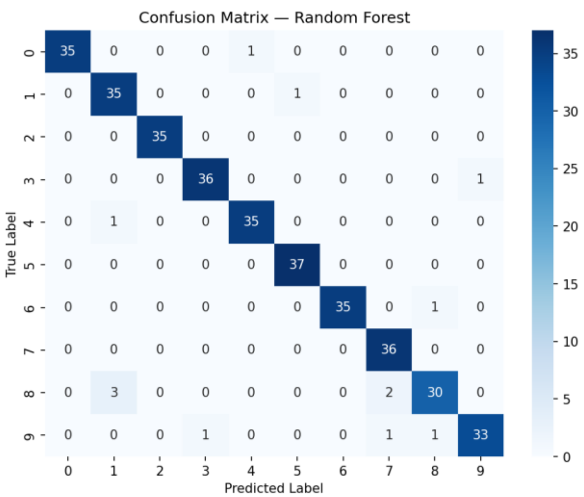

Figure 2. Confusion matrix — Random Forest.

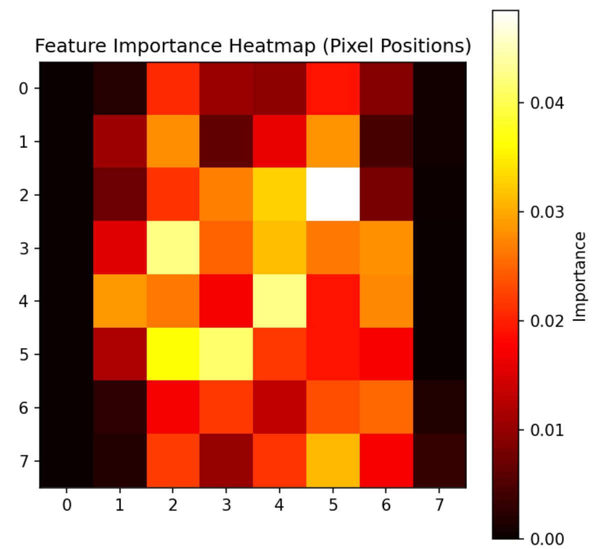

Figure 3. Feature importance by pixel position.

The diagonal dominates the confusion matrix, confirming strong overall performance, but digit 8 is the model's weakest class: 3 instances were misread as 1 and 2 as 7, leaving only 30 of 35 correctly classified. Smaller, isolated confusions also appear between 0/4, 1/5, 6/8, and among 3/7/8/9 — pairs of digits that often share curved or overlapping strokes. The feature importance map shows that corner pixels (e.g., [0,0] and [0,7]) carry almost no weight, since they are background in nearly every sample, while importance concentrates in the central rows and columns: particularly around row 2–col 5 and rows 3–5, cols 1–5: where digit strokes most often pass and differ most between classes. This spatial pattern is a useful sanity check that the model is learning meaningful, human-interpretable structure rather than noise

5. Conclusion

Both models perform strongly on this task, with the SVM achieving the better overall result (97.50% accuracy vs. 96.39% for the Random Forest) and a slightly tighter precision/recall balance. For deployment where raw accuracy is the priority, the SVM is the preferred choice. The Random Forest remains a strong, more interpretable alternative, its feature importance on the map gives direct insight into which pixels drive decisions, and its main source of error. The digit-8 class would be a reasonable target for further feature engineering or additional training data if it is selected instead